ATHARVA AUTI

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EDUCATION

University of Southern California *(M.S. in Cybersecurity Engineering)*, GPA: 3.67/4.0 *Aug 2024 – Aug 2026*

- **Relevant Coursework: INF 519** Foundations & Policy for Information Security, **CSCI 530** Security Systems, **INF 523** Computer Systems Assurance, **ITP 475** Advanced Digital Forensics, **INF 529** Security & Privacy, **INF 517** Research Methods for User Studies

Mumbai University *(B.E. in Cybersecurity, Minors in AI & ML)*, GPA: 9.51/10 *Aug 2020 - Jun 2024*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, OOP Systems, Computer Architecture, Cryptography, Database Management & Security, Web Application Security, Network Security, Virtualization & Cloud Computing, Cyber Laws

EXPERIENCE

Ernst & Young (EY) U.S. LLP *(Cybersecurity Consultant Intern)*, Los Angeles, CA *Jun 2025 – Aug 2025*

- Automated vulnerability data processing for a **Fortune 100** retail client using Python to improve **PowerBI reporting** precision
- Collaborated with client **leadership** & cross-functional technology teams to assess, prioritize, and resolve high-risk vulnerabilities
- Enhanced remediation workflows through **data validation** & **strategic insights** for efficient vulnerability resolution processes
- Ensured regulatory **compliance** for the client through alignment with the **enterprise risk** frameworks and the industry standards

Veermata Jijabai Technological Institute (VJTI) COE CNDS *(Cybersecurity Researcher)*, India *Jun 2023 – Jul 2024*

- Engineered scalable **OT** & **IT infrastructure** leveraging **ESXi** for seamless **SIEM**-capable Security Operations (SOC) deployments
- Integrated **Elasticsearch (ELK)** for data aggregation from multiple **IDS platforms**, supporting in-depth comparative threat analysis
- Strengthened **network security** posture by configuring controls across diverse hardware, including Cisco, FortiGate, and Sophos
- Conducted **penetration testing** exercises using **MITRE ICS,** working on **Threat Modeling** & **Cyber Threat Intelligence (CTI)**
- Spearheaded **Android** application security research for **reverse engineering**, utilizing **Binary Ninja, GDB, IDA Pro,** and **Ghidra**
- Analyzed APK files & **ARM assembly code**, to identify vulnerabilities by analyzing **decompiled code & runtime patterns**

Cyberpeace Foundation, SAKEC (*Cybersecurity Researcher)*, India *Oct 2022 – May 2023*

- Led research initiatives by identifying gaps in security, devising cybersecurity **strategies**, and presenting **findings**
- Developed & deployed a **Threat Intelligence Platform** integrated with **open-source** honeypot technologies (Snare, Tanner, Dionaea)
- Simulated **vulnerabilities**, monitored attacker **behaviors**, & gathered **threat data** using a honeypot in the **educational** sector

PROJECTS

HoneyTrack – Lead for Engineering Project (auti.dev/honeytrack) – Mumbai University *2021 – 2024*

- Devised a Honeypot cum **SIEM** tool operating **Python & BASH** scripts, containerized using **Docker** to automate deployment in **Azure**
- Configured **ELK** for visualization, handling over **20M logs** from 2 months of cloud attack monitoring & improving analysis capabilities
- **Copyrighted** (2022) & published at **IEEE** SCEECS 2023 & Scopus **Journal** of Emerging Technologies & Innovative Research 2023

MobSecOps – An Android Security Framework (auti.dev/mobsecops) – IIT Delhi *2024*

- Enhanced the Mobile Security Framework (MobSF) by integrating modules for static analysis aligned with the new **OWASP** standards
- Ensured comprehensive assessments of **application components, permissions,** and **native libraries**
- Developed advanced **dynamic** analysis features, incorporating **FRIDA scripts** for runtime monitoring, **SSL Certificate Pinning** detection, **root/jailbreak** detection, and **code tampering** analysis to evaluate Android application behavior in real-time
- Integrated the **Llama 3 generative AI** model using **HuggingFace** to get summarized insights by analyzing APK files dynamically
- Proactively generated patterns and adapted to evolving attack vectors for enhanced defense against emerging mobile threats

Asset Management for Operational Technology with SIEM & SOAR (auti.dev/elastic) – VJTI *2023 – 2024*

- Devised an asset management tool for **Cyber-Physical** environments, consisting of Operational Technology systems
- Enabled identification, classification, and real-time monitoring of assets in compliance with **IEC 61850** standards (**MMS** & **GOOSE**)
- Integrated **CVSS**-based vulnerability lookup, endpoint service identification, device **health/liveliness checkup, threat monitoring**, and **live network & PCAP**-based (packet capture) analysis to augment **SOC capabilities** and decision-making
- Enhanced data processing efficiency and **scalability** by implementing **chunking** & **threading** techniques
- Ensured quick handling & analysis of capture files over 100 GB in complex industrial systems with a data center environment

SKILLS & CERTIFICATIONS

- **Skills:** Penetration Testing, SCADA & Cyber-Physical Systems, Cyber Threat Intelligence, Analytics, Risk & Compliance, Network Traffic & Log Analysis, Network Design, Cloud Security, SOC Analysis, Presentation, Consulting
- **Languages & Tools:** Microsoft Word, Excel, PowerPoint, PowerBI, Python, Java, C, C++, PowerShell, BASH, Git, VMware ESXi, Proxmox, Windows Active Directory, Docker, Kubernetes, Azure, GCP, AWS, MySQL, PostgreSQL, FastAPI, Elasticsearch, Kibana, Logstash, Filebeat, Splunk, EZTools, EnCase, KAPE, Autopsy, Volatility, Metasploit, Tenable
- **Frameworks:** ISO/IEC 27001, SOC 2, HIPAA, PCI DSS, CCPA, GDPR, MITRE, Lockheed Martin CKC, CTF (CIA), CIS, CSA, CCM
- **Certifications:** ISC2 CC, EY, Cisco – Networking Essentials, Cloud Security Alliance, Zscaler Workshops, Palo Alto – Cloud, Network Security, and SOC, IBM Cybersecurity & Blockchain, Basel OSINT, Credits for Cryptography & CISSP by Charles Sturt University

ACHIEVEMENTS & LEADERSHIP

- Secured **17th** (4800+ teams) with USC CTF Team & **123rd** (8000+ individuals) in **National Cyber League** *Fall 2024* across the U.S.
- **1st position** at IIT Delhi's Cybersecurity Hackathon *2024* (Cython) against 500+ teams – **Android Security Framework with GenAI**
- **Runner-up** at AI LA Cerebral Beach **Hackathon** *2024* – **Blockchain Smart Contract Auditing using Multiagent AI**
- Presented a **talk** at the Elasticsearch Community Event on **Leveraging Cybersecurity using ELK & developing a SIEM tool**